Two resolutions over 46 seconds

What the extended 256^3 run settles, and the one thing it unsettles

Report II — companion to `report_rot192_rot256`

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Abstract

The first report compared the 192^3 and 256^3 runs at $t = 12\,\mathrm{s}$. The 256^3 run has since been carried to $t = 78\,\mathrm{s}$ and the 192^3 to $60\,\mathrm{s}$, so the comparison spans 48 seconds rather than three and the finer grid runs 18 seconds beyond it. Three results follow. **The braking of L_z converges**: three successive regimes over the shared baseline, each about half the rate of the one before, agreeing between the grids to 7, 7 and 19 percent — so the identification of the angular-momentum transport as magnetic, previously resting on one resolution, now rests on two. Past $60\,\mathrm{s}$ the 256^3 shows a *fourth* regime at $-0.037\%\,\mathrm{s}^{-1}$: the braking does not settle at the third rate, which is what the shorter baseline had suggested. **The steepening of the differential rotation converges, and the longer baseline improves it**: 22.6% against 20.3% to $58.2\,\mathrm{s}$, but 24.3% against 25.4% once each grid is taken to the end of its own data — from 11% apart to 4.5%. **The residual magnetic field does not**: it settles a factor of 25 higher on the finer grid and is still rising at $t = 78\,\mathrm{s}$, 131% above its minimum. A methodological result comes with the second: over a $14\,\mathrm{s}$ window the same two runs appear to disagree about the steepening by a factor of eight, because the 256^3 onset is some $6\,\mathrm{s}$ late. Two earlier drafts drew two different wrong conclusions from that window. **Both convergent results are bounded by the same limitation**, now measured and stated on Report I's first page: the MRI is not resolved, with $Q = \lambda_{\mathrm{MRI}}/\Delta x \simeq 0.4$ over precisely the $t = 40$–$78\,\mathrm{s}$ window in which the steepening is measured. Two grids agreeing does not repair that — both are far below threshold, and neither can host the instability that the resolved literature finds flattening such a profile.

1 What this report adds

Report I established the configuration, the instability and the survival of the star, and compared the two grids over the three seconds where both existed. Its convergence statements were therefore all short-baseline statements. Since then the 256^3 run has been continued through twelve chained windows to $t = 58.2\,\mathrm{s}$ and all 164 plotfiles have been processed with the same diagnostics, so every quantity in Report I can now be asked of both grids over almost the whole evolution.

Nothing about the configuration, the code or the diagnostics has changed. The star is the same $2.005\,M_\odot$ differentially rotating equilibrium with a toroidal-dominated field; the tool is the same `tools/fbtbp.cpp` rebuilding everything from B_x, B_y, B_z.

Throughout, comparisons are made between means over two pulsation periods at each end of the baseline, never between single snapshots. The star breathes by $\pm 14\%$ every $1.498\,\mathrm{s}$, and a snapshot catches an arbitrary phase of it.

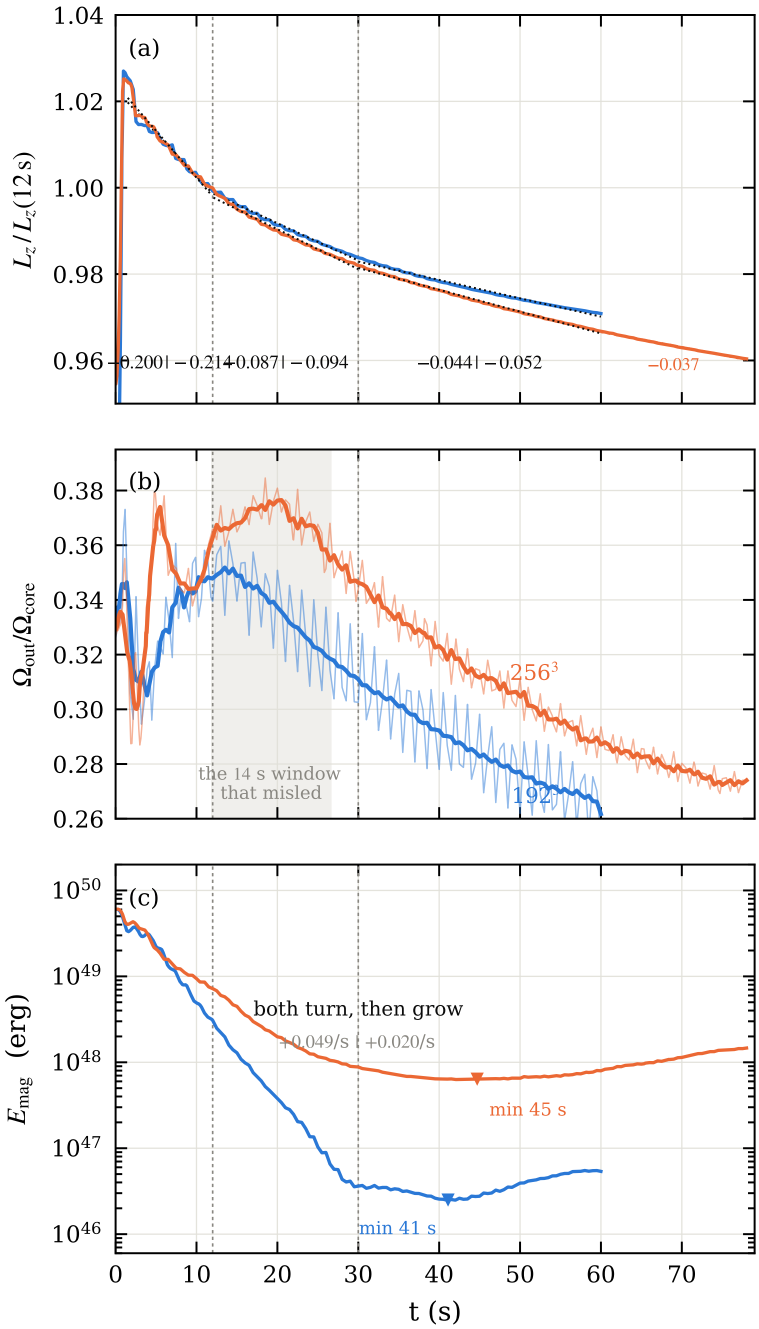

Figure 1: The three results. **(a)** L_z normalised at $t = 12\,\mathrm{s}$, with the three braking regimes fitted; the numbers along the bottom are $192^3 \,|\, 256^3$ in percent per second, and the dashed verticals are the regime boundaries. **(b)** The differential rotation. The finer grid begins to steepen about 6 s late, then steepens faster; the shaded band is the 14 s window over which that lag reads as a disagreement. **(c)** Magnetic energy, the one panel that does not converge. The 192^3 field diagnostics stop at $t = 12\,\mathrm{s}$; the isolated point at $t = 38.5\,\mathrm{s}$ is the single late instant that was processed, and is drawn as a marker because that is all it is.

2 The field, before and after

Before the numbers, the pictures. Fig. 2 is what the ordered field does at 192^3: axisymmetric circles of B_φ in the equatorial cut at $t = 0$, an $m = 1$ deformation growing through $t \simeq 2$–$4\,\mathrm{s}$, filamentation at saturation, and a disordered remnant afterwards. The poloidal row below it starts almost empty — the initial poloidal field is three decades below the toroidal one — and fills in as the instability converts one into the other.

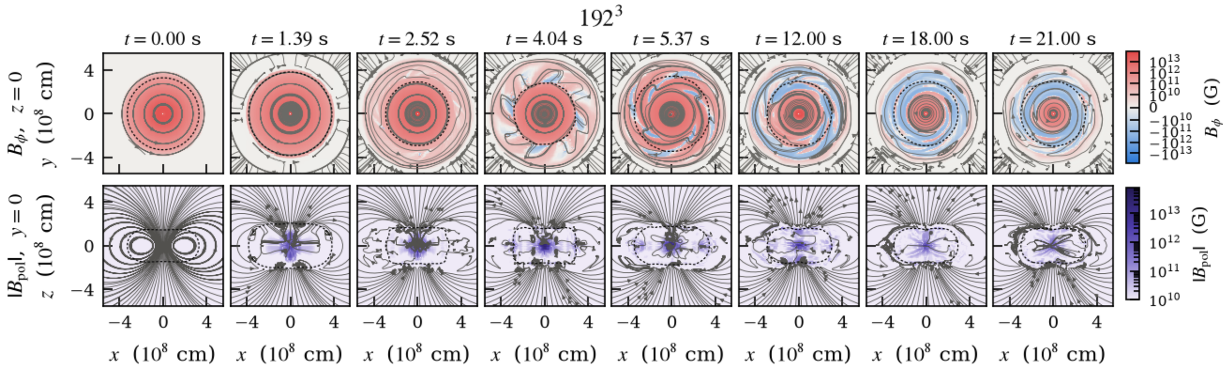

Figure 2: 192^3, $t = 0$ to $21\,\mathrm{s}$. **Top:** equatorial cut $z = 0$, where B_φ lies in the plane, so the streamlines *are* the toroidal field lines — circles while the star is axisymmetric, and their departure from circles is the instability. **Bottom:** meridional cut $y = 0$, where the in-plane field *is* the poloidal one. Colour is B_φ and $|B_{\rm pol}|$, both starting at $10^{10}\,\mathrm{G}$ so the two rows share a threshold in gauss. The dashed contour is $\rho = 10^6\,\mathrm{g\,cm^{-3}}$.

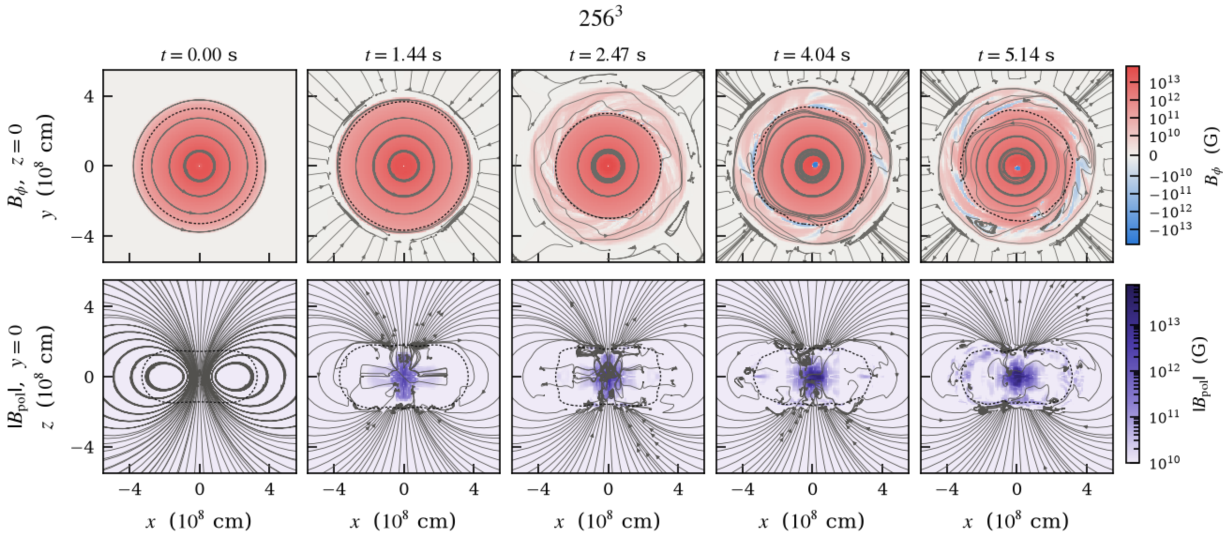

Figure 3: 256^3, same planes, same colour scales, same threshold as Fig. 2 — the two are meant to be compared panel by panel. **This sequence stops at** $t = 5.14\,\mathrm{s}$, the end of the growth phase, because those are the only instants for which slices have been extracted. It therefore covers none of the window this report is about. Extending it is the obvious companion to Sec. 5: a slice at $t \simeq 58\,\mathrm{s}$ would show directly whether the residual field that fails to converge is organised or turbulent, which is the physical question underneath the two readings offered there.

The comparison that matters for this report is not visible in either figure. Both stop before or barely into the window where the two grids disagree about the residual, and the disagreement is a factor of 25 in energy — large enough that it should be plainly visible in a slice, as either a different field strength or a different degree of organisation. That figure is the one piece of evidence this report is missing.

3 The braking of L_z converges

This is the strongest result of the comparison, and it is the one that matters most for the physics, because it is what identifies the transport.

There is no explicit viscosity and no explicit resistivity in these runs, so nothing else in the scheme brakes anything. The star loses angular momentum twice as fast while the ordered toroidal

$\mathrm{d}\ln L_z/\mathrm{d}t$		192^3	256^3	agreement
$t = 1.5$–11.9 s	toroidal field alive	−0.1996%/s	−0.2144%/s	7%
$t = 12$–30 s	field decaying	−0.0873%/s	−0.0935%/s	7%
$t = 30$–58 s	residual	−0.0446%/s	−0.0523%/s	15%

Table 1: Three regimes, each braking at about half the rate of the one before, and the two grids agree on all three.

field exists as it does afterwards, and half as fast again once even the decaying field has gone. That factor of two, repeated, is the signature of magnetic transport, and it is now measured at two resolutions instead of one.

The third regime is new here — Report I described two. With the baseline extended it is clear that the braking does not simply switch from "fast" to "slow" when the field dies but continues to weaken as the field weakens, which is what a magnetic mechanism should do and what a numerical one need not.

What this does not establish. The rates agree between the grids; that does not make them physical. Both runs brake through whatever transport the scheme supplies, and if a component of that is numerical it may well converge at the same rate as the physical part. The argument for the magnetic identification is the *correlation with the field*, not the convergence; convergence only removes the worry that the correlation was a coarse-grid accident.

4 The steepening converges — over a long enough baseline

The differential rotation is the property the configuration depends on: a $2\,M_\odot$ white dwarf exists here only because Ω falls steeply outwards. Report I found that the profile not only survives but steepens, and flagged the amplitude as unconverged.

12–15 s → 55–58.3 s	192^3	256^3	
L_z	−2.6%	−2.9%	converged
$\Omega_{\rm out}$	−15.3%	−14.9%	converged (3%)
$\Omega_{\rm out}/\Omega_{\rm core}$	**−22.6%**	**−20.3%**	**converged** (10%)
$\Omega_{\rm core}$	+9.4%	+6.7%	same sign
$L_z^{\rm outer}/L_z^{\rm inner}$	−10.6%	−6.2%	same sign

Table 2: The steepening over 46 s. The amplitude converges to 10% and the outer-shell spin-down to 3%.

4.1 A convergence test shorter than the lag measures the lag

The amplitude looked irreducibly resolution-dependent over shorter windows, and the reason is worth recording because it is a trap that will recur in this problem.

Neither number is wrong. The short baseline simply does not contain the onset, and a delayed effect measured over a window shorter than its delay is indistinguishable from an absent effect — and, if the lagging curve happens to drift the other way first, from an effect of the opposite sign.

This cost two wrong conclusions in successive drafts of Report I: first that the steepening was a coarse-grid artefact, then that its direction was real but its amplitude was not. Both were stated with the numbers that supported them, and both were wrong for the same reason.

The general form is uncomfortable, because the lag is not known in advance and is not obviously bounded. Here it is 6 s against a 1.5 s pulsation and a 2.4 s Alfvén time — a few dynamical times,

change in $\Omega_{\rm out}/\Omega_{\rm core}$	192^3	256^3
$t = 12 \to 18$ s	−3.2%	**+2.1%**
$t = 18 \to 24$ s	−5.1%	−3.0%
$t = 24 \to 30$ s	−4.5%	**−5.6%**
$t = 30 \to 38$ s	−3.8%	**−5.2%**
total over a 14 s baseline	−8.2%	−1.0%
total over a 46 s baseline	−22.6%	−20.3%

Table 3: The finer grid is still flat — rising, even — while the coarse grid has already begun, and then steepens faster. The last two rows are the same comparison over two baselines: a factor of eight apart at 14 s, ten percent apart at 46 s.

which is reassuring but is a measurement, not a principle. Any future convergence test in this problem needs a baseline long enough that a plausible lag is a small fraction of it.

5 The field stops decaying and regrows — on both grids

This began as the one section where extending the baseline made the disagreement worse. It has become the one positive result in the campaign: something the star *does*, rather than something it loses.

	192^3	256^3	
minimum of $E_{\rm mag}$	$t = 41.1$ s	$t = 44.7$ s	**both turn**
$E_{\rm mag}$ at the minimum (erg)	2.54×10^{46}	6.35×10^{47}	×24.5 apart
growth, whole rise	+0.0485/s	+0.0282/s	*coarse grid faster*
settled rate, $t > 55$ s	—	+0.035/s	e-fold 29 s
minimum to end of run	+114%	+124%	
max $\lvert B\rvert$ over the rise	+45%	+107%	

Table 4: The minima are located by fitting a parabola to $\ln E_{\rm mag}$ over $t = 25$–55 s, so the 1.5 s pulsation does not set the answer. That the turnaround exists is converged; nothing quantitative about it is.

What is converged. Both runs decay, reach a minimum within 4 s of each other, and then grow monotonically to the end of the available data. The mechanism is not exotic and it is available: the differential rotation survives and is still *steepening* (Sec. 4), so there is free shear energy, and winding residual poloidal field into toroidal is the most ordinary thing a shear flow can do to a magnetic field.

The rise is exponential and steady. Carried to $t = 78$ s, the 256^3 growth settles: fitting $\ln E_{\rm mag}$ over $t = 55$–65 gives $+0.0354\,{\rm s}^{-1}$ and over $t = 65$–78 gives $+0.0349\,{\rm s}^{-1}$. Two independent ten-second windows agreeing to 1.5% is an exponential, not a transient, with an e-folding of 29 s. Over the whole rise $E_{\rm mag}$ more than doubles and the peak field goes from 4.05 to 8.37×10^{12} G.

An oscillation would now have to be implausibly slow. The rise has been monotone for 33 s. A sinusoid accelerating away from its minimum inflects at a quarter period, so explaining this needs $P > 133$ s — 280 dynamical times, or 89 of the star's own pulsation periods. Without a named mode at that period the reading is hard to sustain.

The two grids are converging. The gap in $E_{\rm mag}$ narrows monotonically: 24.5 at the minimum, 16.1 at $t = 51$ s, 14.1 at $t = 58$ s, which is as far as the 192^3 run goes. A level set by the mesh would hold its ratio; a level the two runs are both climbing towards would close, and closing is what is observed. Extrapolating the two rates naively has them meeting near $t \simeq 190\,\mathrm{s}$, far beyond either run.

What is still not converged. The rate itself. Over the window both grids cover, the coarse one grows about 2.4 times faster — the same direction, and a similar factor, as the decay rate before it, which fell from 0.282 to $0.141\,\mathrm{s}^{-1}$ under the same refinement. So the phenomenon is robust and its timescale is not.

A prediction that failed, recorded because it was registered. Before this data existed it was written down that the 192^3 would turn *later* and *more weakly*, on the reasoning that more numerical dissipation should delay the point where regeneration overtakes decay. It turns 3.6 s **earlier** and grows 2.5 times **faster**. The prediction was wrong in both directions. What it was testing — whether the turnaround appears at all on the coarse grid — came out yes, which is why the section above survives.

The failure is informative about the mechanism. If the regrowth were a dynamo running against numerical dissipation, less dissipation should mean earlier and stronger, and the opposite is observed. A reading consistent with the numbers is that both grids are winding toward a saturation level, and the one starting 25 times lower has further to travel and so climbs faster in relative terms — but the gap only narrows from 25 to 16 over the available window, so they are not visibly converging on a common level either.

6 What changes in Report I

1. **Strengthened.** The magnetic identification of the transport. Two grids, three regimes, agreement of 7–15%.
2. **Strengthened, and more so with the longer baseline.** The steepening of the differential rotation: 11% apart at 58.2 s, 4.5% apart when each grid is read to the end of its own data. Longer baselines have improved this agreement every time it has been remeasured.
3. **New.** A fourth braking regime past 60 s, at $-0.037\%\,\mathrm{s}^{-1}$ against -0.052 in the third. The deceleration continues rather than settling, so no rate quoted here should be read as the asymptotic one.
4. **Weakened.** Everything about the end state of the field. The residual is a mesh floor and the destruction factor is an upper bound. Report I's Table 4 should be read with that in mind: its mass, radii and rotation rows are measurements, its field rows are bounds.
5. **New caveat.** Convergence tests in this problem need baselines longer than the inter-grid lag, and the lag is only known afterwards.
6. **Bounded.** Both convergent results. The MRI quality factor is now measured from the volume-typical vertical field over the whole run: $Q \simeq 0.4$ across the window used for the steepening, never above 6.5, never near 15. *Agreement between two unresolved grids is not evidence that the physics is captured*, and this report should not be read as strengthening the rotation result against the MRI objection — only against mesh error at fixed physics.

Status and next steps

192^3: complete, $t = 60\,\mathrm{s}$, field diagnostics processed over the whole run. 256^3: complete at $t = 82.7\,\mathrm{s}$, diagnostics processed to 78 s, and everything in this report is measured over that.

The run ended there rather than at the 100 s target: restarting from the last checkpoint aborted after 37 s with `too many subcycles`, deterministically, and the chain's own guard stopped the sequence.

The 192^3 field data between $t = 12$ and 60 s has since been processed, which is why Fig. 1c carries a curve rather than the single point of the first draft.

What would sharpen the report next is not more physical time but the MRI limitation. Three steps, in cost order, none of which needs a longer run:

1. **Local shearing boxes at this star's shear and magnetic Prandtl number.** In the incompressible approximation — valid here to $v/c_s \sim 4 \times 10^{-4}$ — the acoustic timestep disappears and a hundred orbits costs hours rather than months. This measures whether the MRI transports angular momentum at all in a white dwarf interior, where Pm ~ 0.6 sits near the threshold below which the disc literature finds the turbulence dying.
2. **Closure coefficients** for a mean-field model of the unresolved turbulence, either taken from the published neutron-star calibration or remeasured in step 1 if they do not transfer.
3. **The global run with that closure switched on, at both resolutions.** This is the direct test of the central result: the model predicts the inner profile flattening, and both grids here find it steepening.

A genuinely mixed field — comparable poloidal and toroidal energy rather than the 10^{-7} ratio at $t = 0$ used here — remains the configuration the equilibrium literature does not construct and the one expected to survive. An attempt at it is not reported: the twisted-torus model built for it violated $\beta \geq 1$ in the envelope and the run failed at $t = 0.06$ s.

Inputs, submission scripts, diagnostic tools, data and figure code are under version control in the `WD_MAG` repository. The data behind every number here is `investigations/bt_bp_256_long.csv` (210 rows to $t = 78$ s), `investigations/bt_bp_192_late.csv` and `investigations/rotation_192.csv`.